

KSHAMA R BHATT

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EDUCATION

University of Pennsylvania

Master of Science in Engineering, Computer and Information Science (MSE CIS)

Aug 2026 – May 2028 (Expected)

Philadelphia, PA

B.M.S. College of Engineering

B.E. in Information Science and Engineering; **Grade: 4.0/4.0**

Aug 2020 – Jun 2024

Bengaluru, India

EXPERIENCE

Site Reliability Engineer, IBM

IBM Enterprise Application Service for Java (EASeJ) & IBM MQ on Cloud (MCSP)

Aug 2024 – Present

Bengaluru, India

- Operate production **OpenShift/ROKS clusters** on AWS and IBM Cloud across dev, staging, and prod; own version upgrades, worker-node scaling, and TLS reconciliation; tuned Prometheus/Grafana/Instana stack to cut unplanned downtime **30%** and MTTR **40%**.
- Lead **Sev1/Sev2 incident response**: triage, mitigation, customer comms, and structured **Root Cause Analysis (RCA)** with engineering for permanent fixes.
- Shipped end-to-end **AWS Cost Optimization Platform**: Terraform-deployed auto-tagging Lambda across 3 accounts, CUR→S3→Lambda ETL→**DynamoDB**, IAM-auth **API Gateway** cost-query API, and **Grafana** dashboards with Turbonomic for optimization actions.
- Own **BCDR testing** and **environment lifecycle automation**: provision and deprovision short-lived PoC environments via Terraform + Ansible, validate failover drills, verify backup restore paths.
- Built operational automation in **Ansible** and **Python**: **Slack-native JIRA alerting**, automated **Salesforce Change Request** creation, weekly tagging reconciliation; significant toil reduction.
- Built **Jenkins CI/CD** pipelines and **GitOps (ArgoCD)** workflows for cluster validation, compliance drift checks, multi-region config sync; drove **SOC2 compliance** audits on backup alerting, IAM, Secrets Manager rotation.

Software Engineer Intern, IBM

IBM Watsonx & Offering Cost Calculator

Jan 2024 – Jul 2024

Bengaluru, India

- Tuned **DB2** queries and backend logic for high-traffic API; built **Python** + **React** automation using **WatsonxLLM** to replace manual service management workflows.

Software Engineer Intern, Hewlett Packard Enterprise

Open Source Component Selector

Feb 2023 – May 2023

Remote

- Built a full-stack open-source component selector in **Flask**, **MySQL**, **React**; designed backend schemas for metadata-based search and license filtering.

PROJECTS & RESEARCH

Autonomous Reliability Defenders | MCP, Prometheus, Loki, ArgoCD

2nd Place, IBM Bobathon 2026

- Built an **Agentic AI orchestration platform** (presented at **WIST: Women in Software & Technology**): master agent coordinates specialized worker agents (security, capacity, logs, deployment) for pre-deploy validation and post-deploy incident analysis.
- Wrote custom **Model Context Protocol (MCP) servers** for Loki and Prometheus; integrated ArgoCD, Instana, GitHub MCPs; solved **200K context-window limits** via task decomposition and parallel agent execution.

KubePilot (AI-Powered Kubernetes Reliability Platform) | Kubernetes, Go, Prometheus, Grafana, ArgoCD, Terraform, Next.js

- Cloud-native platform that analyzes cluster health, resource utilization, and **upgrade readiness**; surfaces remediation guidance for pod failures, node instability, misconfigured workloads; reduces drift via **ArgoCD GitOps**.

Reinforcement Learning Portfolio Optimization | TensorFlow, OpenAI Gym, DQN | [GitHub](#)

- Applied **Deep Q-Networks (DQN)** for automated portfolio management in an OpenAI Gym environment; designed reward shaping and replay buffer for stable convergence on multi-asset allocation.

Watsonx.data Product Assistant | Python, Flask, Milvus, IBM Granite LLMs

- **RAG** assistant retrieving technical knowledge from Slack and GitHub; **Milvus** vector store + **Granite** LLMs for context-grounded answers.

QuickAid: Ambulance Booking System | Published in IJMRSETM, Vol. 10 Issue 7 | [Publication](#)

- Co-authored emergency response system reducing dispatch latency; published peer-reviewed paper on architecture and field evaluation.

TECHNICAL SKILLS

Cloud & Infra: AWS (Lambda, S3, DynamoDB, API Gateway, IAM, CUR, EC2, RDS), IBM Cloud, Kubernetes, OpenShift/ROKS, Docker

DevOps & IaC: Terraform, Ansible, Jenkins, ArgoCD (GitOps), GitHub Actions, Helm

Observability & SRE: Prometheus, Grafana, Loki, Instana, Alertmanager, SLO/SLI, Incident Mgmt, BCDR, RCA

AI / ML Systems: Model Context Protocol (MCP), Multi-Agent Orchestration, RAG, Milvus, IBM Granite, TensorFlow, DQN, OpenAI Gym

Languages: Python, Go, Java, C/C++, SQL, Bash, JavaScript/TypeScript

Databases: DynamoDB, IBM DB2, MySQL, MongoDB